Senkron Sıkıştırma Dönüşümü Kullanılarak Motor Görüntü BBA Sınıflandırılması
Motor Imagery BCI Classification Using Synchrosqueezing Transform

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Özetçe*— Sensörler ve bilgisayar teknolojisindeki gelişmeler, Elektronsefalografi (EEG) sinyallerinin kaydedilmesini ve işlenmesini kolaylaştırmıştır. EEG sinyalleri yardımıyla beyin-bilgisayar arayüzlerinin (BBA) elde edilmesi daha pratik ve daha ucuz hale gelmektedir. BBA'ları kullanarak beyin sinyallerinin anlamını analiz etmek son yıllarda popüler ve gelecek vaat eden bir araştırma alanıdır. Bu çalışmada, motor görüntü (MI) EEG sinyallerini algılamak ve sınıflandırmak için yeni bir yöntem öneriyoruz. Kullanılan veri seti, hareketlerin tasavvuruna yönelik denemeleri içermektedir. Denekler EEG sinyallerini kaydederken, her deneyde birkaç kez sağ yumruk ve sol yumruk hareketlerini hayal ettiler. Yaklaşımımızda, her deneme için EEG sinyallerinin zaman-frekans temsil matrislerini elde etmek için sychrosqueezing transform (SST) kullanılmaktadır. Temel bileşen analizi, boyut küçültme ve özellik çıkarma için kullanılmıştır. Son olarak, destek vektör makinesi (DVM) kullanılarak sınıflandırma işlemi gerçekleştirilmiştir. Bu algoritma, doğrudan motor hareketlerle ilgili olan seçilmiş EEG kanallarının her birine uygulanmıştır. SST tabanlı yaklaşımımızın test ve eğitim doğrulukları her kanal için ayrı ayrı %99'un üzerindedir. Ayrıca, SST tabanlı yaklaşımla karşılaştırmak için sürekli dalgacık dönüşümü (CWT) kullanarak sınıflandırma sonuçları elde edilmiştir. Her kanalın sınıflandırma sonuçları, SST tabanlı yaklaşımın umut verici performansını açıkça göstermiştir.

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***Abstract*— Developments in sensors and computer technology have made easier to record and process Electrencephalography (EEG) signals. Obtaining brain-computer interfaces (BCI) with the help of EEG signals is getting more practical and cheaper. Analyzing the meaning of the brain signals by using BCIs is popular and promising research area last years. In this work, we propose a new method to detect and classify the motor imagery (MI) EEG signals. The used dataset contains the trials for the imagination of the movements. While recording the EEG signals, subjects imagined the movements of right fist and left fist several times in each experiments. Our approach employes synchrosqueezing transform (SST) to obtain time-frequency representation matrices of EEG signals for each trial. Principal component analysis has been used for dimension reduction and feature extraction. At the end, classification process has been figured out by using support vector machine (SVM). This algorithm has been applied to each of the selected EEG channels which are directly related to motor movements. Testing and training accuracies of our SST based approach are higher than 99% for each channels separately. Also, we obtained classification results using continuous wavelet transform (CWT) to compare with the SST-based approach. Classification results of each channel have showed the promising performance of SST-based approach obviously.**

Keywords—Electroencephalography, Motor Imagery, Brain-Computer Interface, Synchrosqueezing Transform, Support Vector Machine.

I. INTRODUCTION

The brain-computer interface (BCI) is a technology that enables communication between the brain and the external world. BCI directly reads the electrical signals in the human brain, analyzes their meaning, and converts them into control signals to operate external devices. Various neuroimaging techniques are used in BCI, such as functional magnetic resonance imaging (fMRI), electrocorticography (EcoG), magnetoencephalography (MEG), and electroencephalogram (EEG). EEG has become popular due to its low cost, easy portability, and high temporal resolution. EEG signals contain sensorimotor rhythms (SMRs) [1], slow cortical potentials [2], and steady-state visual evoked potential [3]. SMRs are of high application value and have gained much attention.

Overall, SMRs-based BCIs have the potential to revolutionize human-machine interaction and improve the quality of life for a wide range of individuals. Indeed, SMRs-based BCIs have shown promising results in various applications, including improving the quality of life for individuals with disabilities and providing new modes of human-machine interaction. Music generation [4] and computer game control [5] are just a few examples of SMRs-based BCIs' possibilities for healthy users. More advanced devices such as orthoses [6,7], prostheses [8,9], robotic arms [10,11], and mobile robots [12] have also been successfully controlled using SMRs-based BCIs. These technologies can significantly enhance the daily lives of people with disabilities, giving them more independence and mobility [4].

The process of categorizing motor imagery EEG signals in SMR BCIs has some limitations because MI-EEG signals are non-stationary, non-linear, and have a low signal-to-noise ratio (SNR), as stated by [13]. The Wavelet Transform (WT) is a commonly used method for accurately classifying MI-EEG signals through time-frequency analysis. It is essential because it can provide instant information about the EEG signal. Various WT approaches have been applied, such as Discrete Wavelet Transform (DWT) [14,15], Wavelet Packet Transform (WPT) [16], and Dual-Tree Complex Wavelet Transform (DTCWT) [17] for MI-EEG classification. This study aims to use Synchrosqueezing Transform (SST) [18] to obtain more compelling features and improve the classification performance of MI-EEG signals.

II. Materials and Methods

A. Dataset and Method

The data were recorded from 64 electrodes according to the international 10-10 system and sampled at 160 samples per second. Only the MI trials of 91 subjects selected randomly were used for analysis. The selected trials were separated into left fist MI and right fist MI tasks [19]. Seven channels (from C5 to C6) were selected for analysis, and 4 seconds of data were chosen to unify the data dimension since the imagery tasks were performed around 4 seconds. Three experiments were performed for each subject. Each experiment was tried 14 times. We have selected to use seven channels of EEG recordings for each trial. As a result, we have had a total of (91 subjects x 3 experiments x 14 trials) data for classification. Classification tasks are binary, left fist MI or right fist MI.

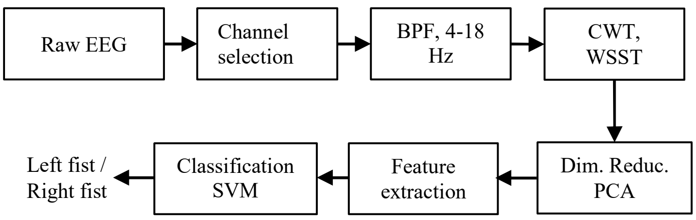

Figure 1. Proposed Motor Image EEG-based BCI Classification System Block Diagram.

As shown in Figure 1, the EEG signals produced with the imagination's left-hand or right-hand movements were selected first. Then the EEG signals were filtered using a 4th-order Butterworth bandpass filter to take information only between 4 Hz and 18 Hz (alpha band). After that, CWT and wavelet-SST (WSST) were employed on the filtered signals separately. Principal Component Analysis (PCA) has been used to reduce the dimensionality of the CWT and WSST matrices. Time-frequency representation matrices of CWT and WSST have been concatenated into a single feature vector using PCA. In the next step, a feature ranking method based on the t-test was employed to select the most effective features for achieving a high classification rate.

B. Synchrosqueezing Transform (SST)

Time-frequency analysis captures nonstationary EEG signals' temporal and frequency characteristics during motor imagery. Traditional time and frequency domain representations cannot capture the sudden temporal variations of the signal while preserving frequency domain information [20]. The SST is a time-frequency method that reassigns the signal energy in frequency and compensates for the spreading effects caused by the mother wavelet [18]. It preserves the time resolution of the signal and can reconstruct an accurate representation of the original signal.

The synchrosqueezing algorithm uses these steps:

1. By obtaining the analytic wavelet, the CWT can capture the instantaneous frequency information of the signal, which is essential for time-frequency analysis. The CWT formula, as given in equation (1), involves convolving the input signal f(t) with a wavelet function ψ((τ-t)/a) that is scaled and shifted in time by the parameters a and t, respectively. The resulting wavelet coefficients $W_f^{\psi}(t,a)$ represent the projection of the input signal onto the wavelet function ψ at a given time and scale.

$$W_f^{\psi}(t,a) = \frac{1}{a}\int f(\tau)\overline{\psi(\frac{\tau-t}{a})}\,d\tau \tag{1}$$

   The scale factor "a" controls the width of the wavelet function in the frequency domain and thus determines the resolution in the time-frequency plane. A smaller value corresponds to a broader wavelet function, which provides better frequency resolution but poorer time resolution. Conversely, a more significant value corresponds to a narrower wavelet function, which provides better time resolution but poorer frequency resolution. The translation parameter "t" determines the centering of the wavelet function in time, thus representing the location of interest in the time domain. Overall, the CWT allows us to analyze the time-varying frequency content of a signal and provides a suitable representation for use with synchrosqueezing analysis.

2. Extract the instantaneous frequencies from the CWT output W_f, by taking the partial derivative of the phase transform with respect to the translation b, divided by the wavelet transform. Therefore, for each scale-time pair (a,b), the instantaneous frequency $w_x(a,b)$ can be evaluated as:

$$w_x(a,b) = -2\pi i W(a,b)^{-1}\frac{\partial W(a,b)}{\partial b} \tag{2}$$

3. The SST method transforms the wavelet coefficients from the scale-time domain to the time-frequency domain. As such, every point (a,b) is transformed into (b, $w_x(a,b)$). Since a and b are discrete values, we need a scaling step. $w_x(a,b)$ was computed only at discrete values a_k, with $\Delta a_k = a_{k-1} - a_k$. During the time-scale to time frequency domain transformation, (b, a) → (b, w_{inst} $(a$, b), the SST coefficient T(w_l, b) is computed only at the centers

w_l of the frequency range $(w_l - \frac{\Delta w}{2}, w_l + \frac{\Delta w}{2})$ with $\Delta w = w_l - w_{l-1}$. Here, w_l is the frequency bins at a resolution of Δw. Here, w_l shows linear frequency scales, so it is a constant value.

In the SST algorithm, the coefficients of the CWT squeezed and reassigned where the phase transform is constant. This reassignment results in a sharpened output from the synchrosqueezing transform when compared to the CWT [18].

The SST of the signal, $T(w_l, b)$, and reconstructed signal, $x(b)$, are given below:

$$T(w_l, b) = \sum_{a_k: |w_x(a_k, b) - w_l| \le \Delta w/2} W(a_k, b) a^{-3/2} \Delta a_k \quad (3)$$

$$x(b) = \mathrm{R}\left[R_\psi^{-1} T(w_l, b) \Delta w\right] \quad (4)$$

$R_\psi = \frac{1}{2}\int_0^\infty \tilde{\psi}^*(\xi)\frac{d\xi}{\xi}$ is the normalization constant and $\hat{\psi}(\xi)$ is the Fourier Transform of the mother wavelet $\psi(t)$.

C. *Feature Extraction and Classification*

Principal Component Analysis (PCA) was employed to reduce the dimensionality of time-frequency images. This technique transformed the data into new features, concatenated into a single vector for each time-frequency image. A feature ranking method based on the Minimum Attainable Classification Error criterion [22-24] for binary classification was used to select the top 500 features.

The Support Vector Machine (SVM) binary classification method aims to find an optimal hyperplane to divide the data into two classes. If the classes are separable, the hyperplane maximizes the margin surrounding it, which acts as a boundary for the positive and negative classes. However, if the classes are inseparable, the algorithm penalizes the margin length for every observation on the wrong side of its class boundary [25].

The SVM score function for linear classification is defined as

$$f(x) = x'\beta + b \quad (5)$$

where x is an observation, β is the vector containing the coefficients that define an orthogonal vector to the hyperplane, and b is the bias term. For separable data, the optimal margin length is 2/‖β‖.

III. Results

Fig.2 illustrates that the CWT features exhibit remarkable classification accuracy during testing. Specifically, channel two yielded the best accuracy and positive predicted values (PPV) of 94.2% and 94.6%, respectively. In comparison, channel seven had a sensitivity of 94.8%, channel two had a specificity of 94.1%, and channel seven had a negative predicted value (NPV) of 94.68%. On the other hand, when testing using WSST features, the highest classification accuracy of 99.5% was achieved in channel five. At the same time, channel five also had a sensitivity, specificity, and NPV of 99.7%. Additionally, channel six had a PPV of 99.6%. Based on these results, it is evident that WSST outperforms CWT.

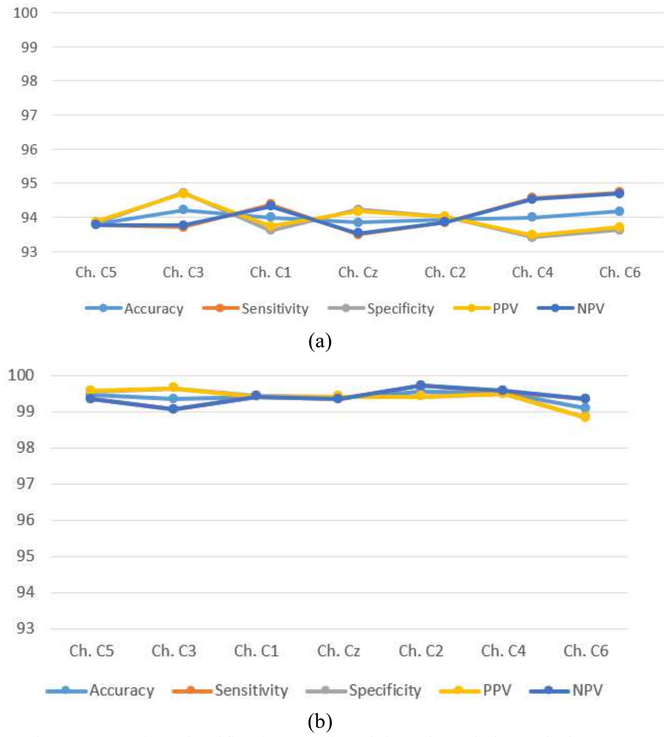

Figure 2. Testing classification results of the selected channels for (a) CWT, and (b) Wavelet-SST.

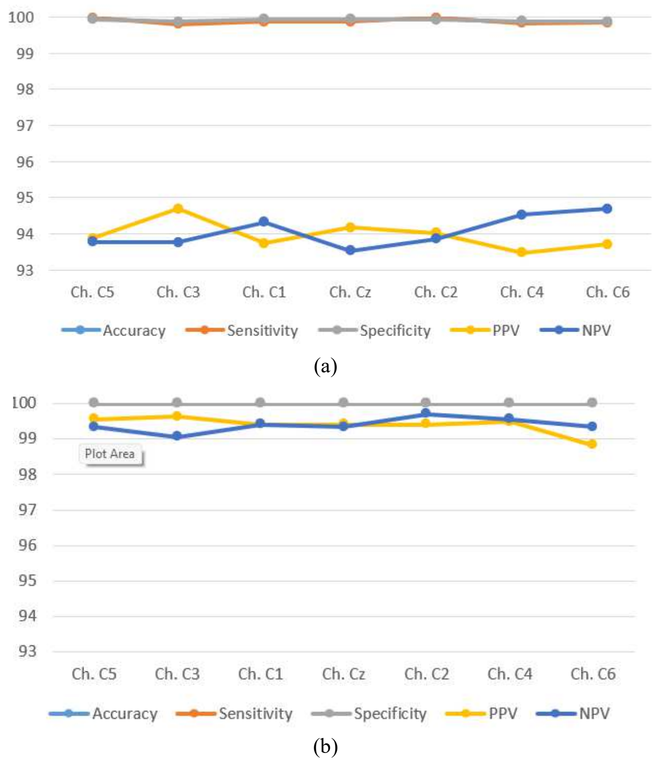

Figure 3. Training classification results of the selected channels for (a) CWT, and (b) Wavelet-SST.

The results presented in Fig. 3 demonstrate the performance of the SVM classifier using WSST and CWT features for training. The WSST features achieved the highest performance, with a negative predicted value of 99.7% in channel five, a positive predicted value of 99.5% in channel two, and 100% classification accuracy, specificity, and sensitivity across all channels. On the other hand, the CWT features yielded a testing classification accuracy and specificity of 99.9% in channel one, a positive predicted value of 94.6% in channel two, a sensitivity of 99.8% in channel four, and a negative predicted value of 94.6% in channel seven. Therefore, the WSST features outperformed the CWT features.

IV. Conclusions

In this study, an SST-based approach has been presented for the classification of MI-EEG signals. At the dimension reduction and feature extraction stage, PCA has been used. We preferred to use SVM for the classification process. All test accuracies are higher than 99%, which is highly satisfactory. It is possible to obtain even better results by trying other machine learning methods. Instead of only imagining right and left fists, more motor movement imaginations like feet and tongue can be added to the classification tasks. Obtaining high accuracies will be more complex in these kinds of classification scenarios. More challenging classification tasks will be tried in future research. As shown in this work, SST performs successfully in analyzing EEG signals and can be used for other classification tasks. By using SST representation of the signal, time and frequency domain information of the signal used together and simultaneously. Using TF representations, more representative and robust features have been extracted. However, for obtaining high classification results, different feature extraction methods and machine learning techniques should be investigated and tried for other MI-EEG classification tasks.

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